

DEPI Assignment 3

Task1:

Create a function called calculateRectangle that: ✓

- Takes two required positional parameters: double length, double width ✓
- Returns a Map with area and perimeter ✓
- Call with length=5, width=3 ✓

Task 2✓

Create a function called createPerson that:✓

- Takes one required String name ✓
- Takes two optional positional parameters: age, city ✓
- Returns a formatted string: "Name: [name], Age: [age or 'unknown'],
City: [city or 'unknown']" ✓
- Call it three ways:
 - createPerson('Guba') ✓
 - createPerson('Ahmed', 22) ✓
 - createPerson('Mohamed', 31, 'Cairo') ✓

Task 3:

Create a function called registerUser that: ✓

- Takes required named parameters: email, password ✓
- Takes optional named parameter: phone ✓
- Returns a confirmation message including all provided info ✓
- Call with email="user@example.com", password="123456" ✓

Task 4:

Create a function called sendMessage that:✓

- Takes required positional: recipient✓
- Takes optional positional: message
- Takes optional named: urgent (with default value false) ✓
- Returns formatted message
- Call: sendMessage('Ahmed', 'Hello!', urgent: true) ✓

Task 5:

Create a function called filterNumbers that:

- Takes List numbers and a bool Function(int) predicate
- Returns filtered list where predicate returns true
- Call with 3 anonymous functions:
 - filters even numbers
 - filters odd numbers
 - filters prime numbers